

Bitcoin Pricing with Heston Model

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1. Introduction & Motivation

- **Context:**

- The Classical Black-Scholes model assumes constant volatility, failing to capture the "Volatility Smile" and clustering observed in real markets.
- **Bitcoin (BTC)** represents a nascent asset class with extreme volatility, skewness, and heavy tails.

- **Objectives:**

- Implement the **Heston Stochastic Volatility Model** to price Bitcoin options.
- Compare two calibration methodologies: **One-Stage (Joint)** vs. **Two-Stage**.
- Benchmark performance against **SPY (S&P 500)** options to highlight structural differences.

2. Mathematical Framework: Heston Dynamics

The model assumes the asset price S_t and variance v_t follow the SDEs under the risk-neutral measure:

Asset Price Dynamics

$$dS_t = (r - q)S_t dt + \sqrt{v_t}S_t dW_t^S \quad (1)$$

Variance Process Dynamics

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^v \quad (2)$$

Parameters:

- κ : Mean reversion speed; θ : Long-term variance.
- σ : Volatility of volatility (Vol-of-Vol).
- ρ : Correlation between dW_t^S and dW_t^v ($dW_t^S \cdot dW_t^v = \rho dt$).

Feller Condition for stability: $2\kappa\theta > \sigma^2$.

2. Pricing via Characteristic Functions

Semi-closed form solution for European Call Options:

$$C(S_0, K, T) = S_0 e^{-qT} P_1 - K e^{-rT} P_2 \quad (3)$$

Probabilities P_j are computed via Fourier Inversion:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-iu \ln K} \phi_j(u; T, S_0)}{iu} \right] du \quad (4)$$

- **Characteristic Function** ϕ_j :

$$\phi_j(u) = \exp(C_j(u, T) + D_j(u, T)v_0 + iu \ln S_0)$$

- **Implementation:** Integration using adaptive quadrature with a dynamic upper bound to ensure stability ("Little Heston Trap" avoidance).

3. Calibration Methodology

We implemented two distinct calibration approaches:

Method 1: One-Stage (Joint)

- Optimizes all 5 parameters ($\kappa, \theta, \sigma, \rho, v_0$) simultaneously.
- Objective: Minimize weighted Sum of Squared Errors (SSE).
- Includes penalty for Feller condition violations.

Method 2: Two-Stage

- **Stage 1:** Fix ρ , calibrate volatility params to match the Implied Volatility Surface.
- **Stage 2:** Fix vol params, calibrate ρ to match prices.
- Goal: Improve stability by reducing dimensionality.

3. Data: Deribit Bitcoin Options

Dataset: Deribit BTC Options (Dec 2025).

- Spot Price $S_0 \approx \$92,497$.
- Market exhibits high baseline volatility (ATM $\approx 55\%$).

Filtering Pipeline:

- 1 **Maturity:** Removed $T < 0.004$ yrs.
- 2 **Liquidity:** Bid Price > 0 .
- 3 **Validity:** Mark Price > 0.0005 BTC.

Final Sample: 469 valid options.

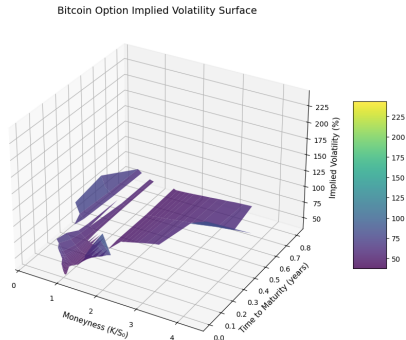


Figure 1: Bitcoin Implied Volatility Surface

4. Results: Calibrated Parameters

Comparison of parameters obtained from the two methods:

Parameter	One-Stage	Two-Stage
κ (Mean Reversion)	2.0164	3.1740
θ (Long-term Var)	0.2945	0.5662
σ (Vol-of-Vol)	1.0898	1.3885
ρ (Correlation)	-0.1220	-0.9390
v_0 (Initial Var)	0.2492	0.6722
Feller Cond.	Marginally Met	Satisfied

Insight: The high $\sigma > 1.0$ confirms the extreme volatility of Bitcoin. Two-stage calibration forces an unrealistic correlation close to -1 .

4. Results: Pricing Performance

Error Metrics (MAPE):

- **One-Stage: 17.3%**
(Superior fit)
- **Two-Stage: 103.8%**
(Severe underperformance)

Analysis:

- One-Stage method provides a stable fit across most strikes.
- Two-Stage method overfits the skew, failing at Deep OTM options.

Heston Calibration Comparison: One-Step vs Two-Step

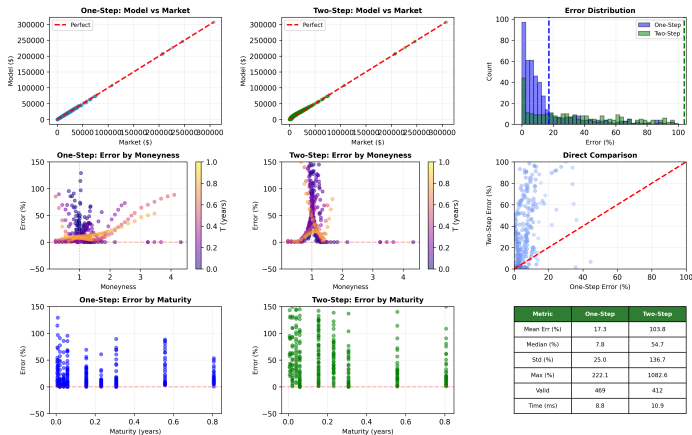


Figure 2: Model vs Market Prices & Errors

4. Results: Volatility Surface Fit

- **Smile Capture:** The Heston model captures the ATM level well but produces a "flatter" smile than observed.
- **Term Structure:** The market shows an inverted term structure. The model struggles to perfectly match this decay rate.

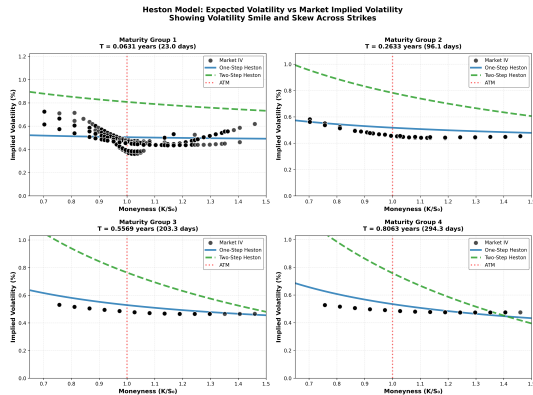


Figure 3: Volatility Smile Fit (Model vs Market)

5. Benchmark: Application to SPY

We applied the same model to **SPY (S&P 500)** options (Dec 2022).

Metric	Value
MAPE Error	3.02%
Fit Quality	Excellent
Feller Cond.	Satisfied

Conclusion: Heston fits mature equities perfectly but struggles with Crypto.

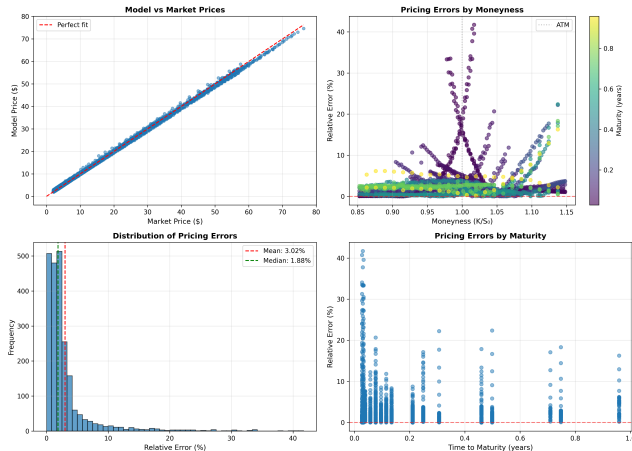


Figure 4: SPY Calibration Results

5. Discussion: Why the Discrepancy?

Why is the error for Bitcoin (17.3%) much higher than SPY (3.0%)?

- ➊ **Asset Characteristics:** SPY follows a continuous diffusion. Bitcoin exhibits **jumps** and extreme kurtosis that standard Heston cannot capture.
- ➋ **Parameter Instability:** Bitcoin's high Vol-of-Vol pushes parameters to the boundary of the Feller condition ($2\kappa\theta \approx \sigma^2$), causing numerical instability.
- ➌ **Data Noise:** Bitcoin options have wider spreads and more noise in the tails compared to the liquid SPY market.

6. Conclusion & Future Work

- **Summary:**

- Implemented Heston model with Fourier pricing.
- One-Stage calibration is the superior method for Bitcoin.

- **Key Takeaway:**

- While Heston works well for equities (SPY), it is **insufficient** to fully capture Bitcoin's "rough" volatility surface.

- **Future Work:**

- Incorporate **Jumps** (Bates Model) or **Rough Volatility** models to better fit the short-term skew.